

GANPAT UNIVERSITY
B. TECH SEM-IV (INFORMATION TECHNOLOGY)
SECOND INTERNAL EXAMINATION – APRIL 2025
2CEIT404: Programming with Python

Exam No: _____

TIME: 1 Hour

TOTAL MARKS: 20

- Instructions:**
- 1) Figures to the right indicate full marks.
 - 2) Be precise and to the point in your answer.
 - 3) The text just below marks indicates the Course Outcomes Numbers, (CO) followed by the bloom's taxonomy level of the question, i.e., R: Remembering, U: Understanding, A: Applying, N: Analyzing, E: Evaluating, C: Creating.

- Q.1** Explain the purpose and syntax of *args and **kwargs in Python functions, and demonstrate their usage with practical examples. How do these special parameters allow for flexibility when defining functions that need to handle varying numbers of arguments? **[5]**
1U
- Q.2** Explain the different file modes available in Python's file handling. Write a program that reads a text file, counts the frequency of each word, and writes the results to a new file in descending order of frequency. **[5]**
2A
- Q.3** Write a Python program that demonstrates the following OOP concepts: **[5]**
2A
- Class and static variables with appropriate examples
 - Constructor and destructor methods
- Q.4** Develop a user-defined module named 'shape_calculator.py' with functions to calculate area and perimeter/circumference of different shapes (rectangle, square, circle, and triangle). Then create a main program that imports this module and allows users to select a shape and input the necessary measurements to calculate the area and perimeter. **[5]**
1C

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